

Bitsmedia: Enhancing the Muslim lifestyle around the globe with Google Cloud

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ABOUT BITSMEDIA

Bitsmedia built a Muslim lifestyle app and achieved exponential growth across 75 countries, handled Ramadan traffic surges, and developed a streaming service with unique features via Google Cloud.



About Bitsmedia

Bitsmedia is a Singapore-based tech company that launched the world's most popular Muslim lifestyle app. Muslim Pro has more than 75 million users in more than 200 countries, offering religious, lifestyle, and community features, including prayer times and halal restaurant guidance. It was awarded the Islamic Economy Award for media category at the Global Islamic Economic Summit 2018.

Industries: Technology

Google Cloud results

- Provides scalability to serve the global Muslim community
- Frees developers to create a stream of community features
- Enables hundreds of millions of daily server requests to be effectively handled
- Ensures maximum security for sensitive religious data
- Inspires team to evolve

Enable growth million every r

Location: Singapore

(<https://cloud.google.com/products/operations>)

App Engine (<https://cloud.google.com/appengine/>)

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud CDN (<https://cloud.google.com/cdn/>)

Firebase (<https://firebase.google.com/>)

Operations (<https://cloud.google.com/products/operations>)

Google Cloud (<https://cloud.google.com/>)

**platform
with ML
solutions**

During the holy month of Ramadan, nearly two billion Muslims around the globe observe a period of prayer and fasting in places as diverse as Indonesia, Morocco, Sweden, and Burkina Faso. Whether the believer is in a massive city or a desert outpost, they must know what time to begin their fast and in which direction to pray toward Mecca.

Singapore-based start-up [Bitsmedia](https://www.bitsmedia.com/)

(<https://www.bitsmedia.com/>) launched its Muslim Pro mobile app to serve the needs of the global Muslim community, from calculating prayer times and the direction to Mecca from anywhere in the world, to helping Muslims learn to read the Quran. As a lifestyle platform, it's constantly adding features, such as halal restaurant recommendations (aiming to become the TripAdvisor of the Muslim world) and a community prayer service it likes to call the "Twitter of compassion."

"With Google Cloud, you just write your code, focus on the application logic, and let Google handle the rest," says Erwan Macé, CEO at Bitsmedia.



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—Erwan Macé, founder,
Bitsmedia

Answering the call of unexpected Muslim needs with Google Cloud

When Bitsmedia released Muslim Pro in 2010, a year after the company’s founding, the app was one of several in a lineup of platforms. Erwan and his team saw it “almost as a pet project we were using to test the Indonesian market, the biggest Muslim population in the world.” He had no idea that the project would enjoy such spectacular growth to become the world’s dominant Muslim lifestyle app.

Word-of-mouth turned the platform into a viral phenomenon, as it struck a chord with digital lifestyle demands in the Muslim community. The market went beyond traditional Muslim countries to places like France, Erwan’s home country, where minority Muslims can benefit from the app’s guidance on things like the location of their nearest mosque and where to find halal food. Within two years of its launch, Muslim Pro reached one million users. From there, it rapidly reached two, five, 15, and eventually 75 million people around the world: “Now we’re gaining a million users every 30 days,” says Erwan.

Supporting this exponential scale curve was the virtually unlimited server capacity of Google Cloud (<https://cloud.google.com/>): “Google Cloud allows me to stop worrying about how the platform is going to scale,” says Erwan. “It’s the only way we can sustain such a huge volume of traffic. We have up to four million daily active users, which means hundreds of millions of server requests per day.”

The computing challenge lay not only in the number of users, but in their distribution over every continent. For Bitsmedia, the Google Cloud network of data centers in 20 regions, 61 zones, and more than 200 countries guarantees reliable connections to the Muslim Pro app, wherever users happen to be.

But it was Ramadan that demonstrated just how powerful Google Cloud could be in enabling Bitsmedia to serve the needs of the world’s Muslims. If the seasonal peak for ecommerce sites is Christmas or Black Friday, for Muslim Pro it’s Ramadan. The traffic “goes through the roof,” as Erwan says, in the first week of the fasting period, surging from about four million daily users to roughly 10 million.

“That’s when Google Cloud makes perfect sense,” says Erwan. “If we had our own infrastructure, we would have to find ways to sustain the Ramadan traffic and buy sufficient machines to support something that happens once a year. But with Google Cloud, we can instantly scale up or down whenever we need to.”

Unleashing innovation with powerful development tools

Bitsmedia's strong growth is powered by its ability to develop features that attract new users to its platform. It recently added an ecommerce shop to Muslim Pro that offers products such as halal cosmetics, Islam-related children's books, prayer mats, and head-covering fashionwear.

Enabling the constant flow of features is the Google Cloud ecosystem of services, which spans computing power, app development, and data analytics. In particular, Erwan's team relies on a trio of Google Cloud tools: [App Engine](https://cloud.google.com/appengine/) (<https://cloud.google.com/appengine/>), [Firebase](https://firebase.google.com/) (<https://firebase.google.com/>), and [BigQuery](https://cloud.google.com/bigquery/) (<https://cloud.google.com/bigquery/>), which allows it to focus on building the app, deploying it, and testing its success.

App Engine is the core development platform, while Firebase accelerates the team's ability to deploy rapidly to the mobile app. Meanwhile, BigQuery allows Bitsmedia to measure user engagement with each new feature, allowing it to nurture popular ones and drop those that don't gain traction.

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We waste less
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—Erwan Macé, founder,
Bitsmedia

“With Google Cloud, we’re constantly rolling out new features,” says Erwan. “On average we go live with one major new feature every three or four months, and one small feature every month.”

Firebase also played a particularly strong role in the Bitsmedia journey. Ahmadul Hoq, Bitsmedia’s Head of Mobile Engineering, recalls one lunch with team members that turned into an ideation session about creating a mobile community for the Hajj pilgrimage to Mecca: “Because of Firebase, we managed to

develop the full version of the feature in less than a week.”

“Firebase plays the biggest role in allowing us to deploy things very fast,” Erwan adds. “Modules such as real-time database and authentication are almost just plug-and-play. We waste less time in writing the back end by using ready-made features.”

Ultimate security for sensitive personal data

As a Muslim lifestyle app, Bitsmedia’s platform handles a great deal of personal data, all the more sensitive because much of it concerns religious preferences. The robust security model of Google Cloud provides the framework necessary to keep the religious data of 75 million users around the world safe.

“We couldn’t do it on our own,” says Erwan. “Only a platform like Google Cloud can provide that level of security and scalability.”

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Limitless future in building global community platform

Next, Bitsmedia plans to leverage the machine learning capabilities of BigQuery to better personalize the user experience on Muslim Pro. It envisages Quranic passage recommendations based on verses users have read in the past. And it plans to use data to craft tailor-made messaging to visitors on its ecommerce shop, directing them to the most relevant products based on their browsing history.

Most of all, Google Cloud and its machine learning solutions are inspiring the Bitsmedia team to find limitless possibilities for experimentation and identify new ways to enrich Muslim lives around the world. “We don’t always know what we want to make,” says Ahmadul. “But Google Cloud machine

learning tools inspire me to create new solutions in this area and fill me with hope for building a better future.”

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Google Cloud
can help your
business.

